Adarsh Sahu

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SUMMARY

Undergraduate student at NIT Jamshedpur with hands-on experience in machine learning, analytics, anomaly detection, forecasting, and RAG-based AI applications. Built end-to-end projects using Python, SQL, pandas, scikit-learn, Streamlit, FAISS, and LLM APIs, with a focus on practical, data-driven applications and interactive deployment.

EDUCATION

National Institute of Technology Jamshedpur — **Jamshedpur, Jharkhand**
Bachelor of Technology (B.Tech) | **CGPA: 7.45/10** — **2024 – 2028**

PROJECTS

AI Study Buddy | ***Python, Streamlit, Groq API, RAG, FAISS, sentence-transformers, PyMuPDF*** **2026**

- Built an exam-focused RAG application that converts handwritten and typed PDF notes into a searchable knowledge base for question answering and structured revision support.
- Implemented an end-to-end pipeline with document parsing, chunking, embedding generation, FAISS vector search, and retrieval-grounded response generation over uploaded study material.
- Added previous-year-question analysis to identify recurring topics and generate targeted summaries, important questions, and last-minute revision content for faster exam preparation.

IPL 2026 Winner Prediction | ***Python, pandas, scikit-learn, XGBoost, Streamlit, Web Scraping*** **2026**

- Developed a cricket forecasting system using XGBoost to estimate team win probabilities from historical IPL match data, current season trends, and simulation-based projections.
- Engineered match-level features including recent form, venue advantage, head-to-head record, and team balance from historical and live-updated cricket datasets.
- Built a Streamlit dashboard to display predictions, team comparisons, points-table context, and season-outcome insights in an interactive format.

Customer Churn Prediction | ***Python, pandas, NumPy, scikit-learn, Streamlit, Seaborn*** **2026**

- Built an end-to-end churn prediction system on the IBM Telco Customer Churn dataset, covering data cleaning, EDA, preprocessing, model training, and deployment on 7,032 customer records.
- Created reusable preprocessing workflows using Pipeline and ColumnTransformer with scaling, one-hot encoding, and feature preparation for robust training.
- Compared Logistic Regression and XGBoost, selected the stronger model based on accuracy and ROC-AUC, and deployed a Streamlit app achieving about 80% accuracy and 0.84 ROC-AUC.

Stock Anomaly Detector | ***Python, pandas, scikit-learn, yfinance, Streamlit, Matplotlib*** **2026**

- Developed a market monitoring app using Isolation Forest to detect unusual price and volume behavior in NSE-listed stocks from multi-year OHLCV data.
- Engineered anomaly features such as daily return, intraday range, and volume ratio to identify abnormal market activity without labeled event data.
- Built an interactive Streamlit dashboard to visualize anomalous trading periods and support event-based investigation of unusual stock movements.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML/CSS

Data Analytics: EDA, data cleaning, feature engineering, data visualization, statistical analysis, forecasting

Machine Learning: regression, classification, anomaly detection, model evaluation, preprocessing pipelines, time-series forecasting, simulation-based prediction

Libraries / Tools: pandas, NumPy, Matplotlib, Seaborn, scikit-learn, XGBoost, Streamlit, Jupyter Notebook, Git, GitHub, VS Code

AI / LLM Concepts: RAG, embeddings, vector databases, FAISS, prompt engineering, LLM application development

Currently Learning – ML Foundations: probability and statistics, linear algebra, calculus, hypothesis testing, bias–variance tradeoff, regularization, cross-validation, hyperparameter tuning, imbalanced-data handling

Currently Learning – Deep Learning: neural networks, backpropagation, activation and loss functions, TensorFlow, PyTorch, transformers, attention

Currently Learning – ML Engineering: modular and testable Python, FastAPI, Flask, Docker, AWS deployment, model monitoring, experiment tracking